

Table 2

H1 dataset summary statistics – Date variables.

Variable	Min	Max	Median	Unique
<i>ReservationStatusDate</i>	2014-11-18	2017-09-14	2016-07-31	913

Table 3

H1 dataset summary statistics – Categorical variables.

Variable	Unique	Top counts
<i>Agent</i>	186	240: 13 095, NULL: 8 209, 250: 2 869, 241: 1 721
<i>ArrivalDateMonth</i>	12	Aug: 4 894, Jul: 4 573, Apr: 3 609, May: 3 559
<i>AssignedRoomType</i>	11	A: 17 046, D: 10 339, E: 5 638, C: 2 214
<i>Company</i>	236	NULL: 36 952, 223: 784, 281: 138, 154: 133
<i>Country</i>	125	PRT: 17 630, GBR: 6 814, ESP: 3 957, IRL: 2 166
<i>CustomerType</i>	4	Tra.: 30 209, Tra.-Party: 7 791, Con.: 1 776, Gro.: 284
<i>DepositType</i>	3	No Dep.: 38 199, Non-Refund.: 1 719, Ref.: 142
<i>DistributionChannel</i>	4	TA/TO: 28 295, Dir.: 7 865, Cor.: 3 269, Und.: 1
<i>IsCanceled</i>	2	0: 28 938, 1: 11 122
<i>IsRepeatedGuest</i>	2	0: 38 282, 1: 1 778
<i>MarketSegment</i>	6	Onl.: 17 729, Off.: 7472, Dir.: 6 513, Gro.: 5 836
<i>Meal</i>	5	BB: 30 005, HB: 8 046, Und.: 1 169, FB: 754
<i>ReservationStatus</i>	3	C.Out: 28 938, Can.: 10 831, No-Show: 291
<i>ReservedRoomType</i>	10	A: 23 399, D: 7 433, E: 4 892, G: 1610

Table 4

H1 dataset summary statistics – Integer and numeric variables.

Variable	Mean	SD	P0	P25	Median	P75	P100
<i>ADR</i>	94.95	61.44	-6.38	50	75	125	508
<i>Adults</i>	1.87	0.7	0	2	2	2	55
<i>ArrivalDateOfMonth</i>	15.82	8.88	1	8	16	24	31
<i>ArrivalDateWeekNumber</i>	27.14	14.01	1	16	28	38	53
<i>ArrivalDateYear</i>	2016.12	0.72	2015	2016	2016	2017	2017
<i>Babies</i>	0.014	0.12	0	0	0	0	2
<i>BookingChanges</i>	0.29	0.73	0	0	0	0	17
<i>Children</i>	0.13	0.45	0	0	0	0	10
<i>DaysInWaitingList</i>	0.53	7.43	0	0	0	0	185
<i>LeadTime</i>	92.68	97.29	0	10	57	155	737
<i>PreviousBookingsNotCanceled</i>	0.15	1	0	0	0	0	30
<i>PreviousCancellations</i>	0.1	1.34	0	0	0	0	26
<i>RequiredCarParkingSpaces</i>	0.14	0.35	0	0	0	0	8
<i>StaysInWeekendNights</i>	1.19	1.15	0	0	1	2	19
<i>StaysInWeekNights</i>	3.13	2.46	0	1	3	5	50
<i>TotalOfSpecialRequests</i>	0.62	0.81	0	0	0	1	5

contains 100 observations. The bars in each variable show the mean value for numeric variables or the frequency of each level for categorical variables. Analyzing these figures it is possible to verify that, for both of the hotels, the distribution of variables like *Adults*, *Children*, *StaysInWeekendNights*, *StaysInWeekNights*, *Meal*, *Country* and *AssignedRoomType* is clearly different between non-canceled and canceled bookings.

Table 5

H2 dataset summary statistics – Date variables.

Variable	Min	Max	Median	Unique
<i>ReservationStatusDate</i>	2014-10-17	2017-09-07	2016-08-10	864

Table 6

H2 dataset summary statistics – Categorical variables.

Variable	Unique	Top counts
<i>Agent</i>	224	9: 31 955, NULL: 8 131, 1: 7 137, 14: 3 640
<i>ArrivalDateMonth</i>	12	Aug: 8 983, May: 8 232, Jul: 8 088, Jun: 7 894
<i>AssignedRoomType</i>	9	A: 57 007, D: 14 983, E: 2 168, F: 2 018
<i>Company</i>	208	NULL: 75 641, 40: 924, 67: 267, 45: 250
<i>Country</i>	166	PRT: 30 960, FRA: 8 804, DEU: 6 084, GBR: 5315
<i>CustomerType</i>	4	Tra.:59 404, Tra.-P.: 17 333, Con.: 2 300, Gro.:293
<i>DepositType</i>	3	No Dep.: 66 442, Non-Refund.: 12 868, Ref.: 20
<i>DistributionChannel</i>	5	TA/TO: 68 945, Dir.: 6 780, Cor.: 3 408, GDS: 193
<i>IsCanceled</i>	2	0: 46 228, 1: 33 102
<i>IsRepeatedGuest</i>	2	0: 77 298, 1: 2 032
<i>MarketSegment</i>	8	Onl.: 38 748, Off: 16 747, Gro.: 13 975, Dir.: 6 093
<i>Meal</i>	4	BB: 62 305, SC: 10 564, HB: 6 417, FB: 44
<i>ReservationStatus</i>	3	C.Out: 46 228, Can.: 32 186, No-Show: 916
<i>ReservedRoomType</i>	8	A: 62 595, D: 11768, F: 1 791, E: 1 553

Table 7

H2 dataset summary statistics – Integer and numeric variables.

Variable	Mean	SD	P0	P25	Median	P75	P100
<i>ADR</i>	105.3	43.6	0	79.2	99.9	126	5400
<i>Adults</i>	1.85	0.51	0	2	2	2	4
<i>ArrivalDateOfMonth</i>	15.79	8.73	1	8	16	23	31
<i>ArrivalDateWeekNumber</i>	27.18	13.4	1	17	27	38	53
<i>ArrivalDateYear</i>	2016.17	0.7	2015	2016	2016	2017	2017
<i>Babies</i>	0.0049	0.084	0	0	0	0	10
<i>BookingChanges</i>	0.19	0.61	0	0	0	0	21
<i>Children</i>	0.091	0.37	0	0	0	0	3
<i>DaysInWaitingList</i>	3.23	20.87	0	0	0	0	391
<i>LeadTime</i>	109.74	110.95	0	23	74	163	629
<i>PreviousBookingsNotCanceled</i>	0.13	1.69	0	0	0	0	72
<i>PreviousCancellations</i>	0.08	0.42	0	0	0	0	32
<i>RequiredCarParkingSpaces</i>	0.024	0.15	0	0	0	0	3
<i>StaysInWeekendNights</i>	0.8	0.89	0	0	1	2	16
<i>StaysInWeekNights</i>	2.18	1.46	0	1	2	3	41
<i>TotalOfSpecialRequests</i>	0.55	0.78	0	0	0	1	5

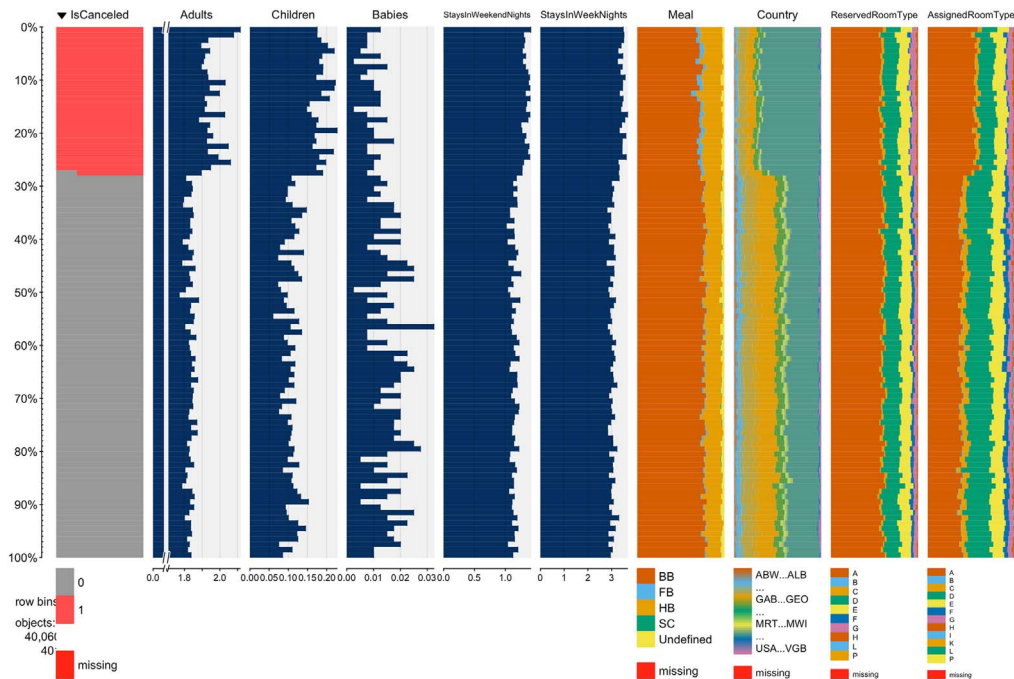


Fig. 2. H1 dataset partial visualization of all observations.

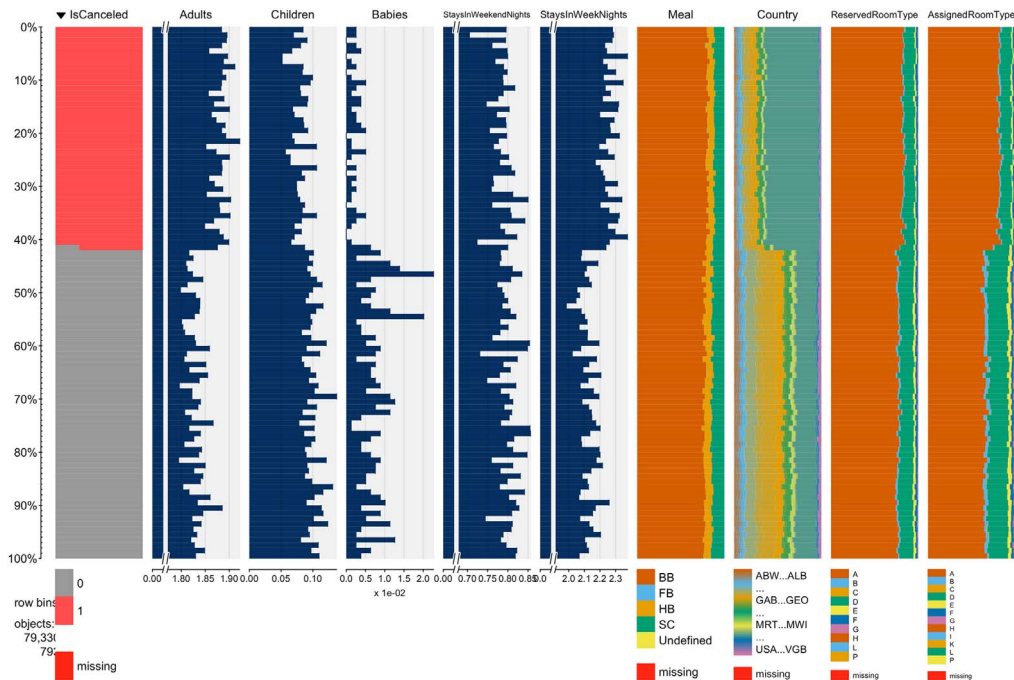


Fig. 3. H2 dataset partial visualization of all observations.